

Predicting F1 Tyre Degradation to Optimize Pit-Stop Strategy

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Kevin	data pipeline, cleaning, feature engineering and target iteration, modeling pipeline, pit-stop optimizer, back-test framework, presentation
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1. Why this problem

In Formula 1, tyre strategy directly determines race outcomes. A pit stop costs roughly 25 seconds of track time, so the decision of when to stop and which compound to switch to is one of the most consequential calls a race engineer makes. Stop too early and you pay for an extra pit visit; stop too late and the tyres lose grip catastrophically over the closing laps. Teams currently combine proprietary simulators, real-time telemetry, and engineer judgment to make these calls, but no tool exists for enthusiasts or students to analyze race data and make their own strategy. We wanted to see whether we could build a tool using only public data and standard machine learning techniques. Our success metric was simple: a strategist should be able to glance at the tool's recommendation and say "yeah, that's a sensible call."

2. Dataset and preprocessing

All data comes from the FastF1 Python library, which provides official F1 timing data. We pulled the full 2024 and 2025 seasons. That gave us 48 race sessions across 24 unique circuits, yielding around 53,000 raw laps. Each row contains lap time, tyre compound, tyre life (laps on the current set), stint number, pit events, and weather data.

Cleaning was aggressive because we wanted a dataset that reflects only genuine on-pace tyre wear. We removed in and out laps (pit-lane speed limits distort timing), safety car / virtual safety car (VSC) / red-flag laps (speed is artificially capped), wet and intermediate compounds (out of scope for our dry-only model), the lap-1 standing start (cold-tyre chaos rather than wear), laps flagged as inaccurate by the FastF1 timing system, and statistical outliers more than 3 standard deviations from each race's mean (typically spins or collisions). We also dropped rows missing key fields. The result is 43,319 clean laps, an 18.7% reduction from the raw set.

The most difficult data-prep decision was choosing an appropriate modeling target. Our initial choices failed: subtracting the per-stint median absorbed the wear signal we

wanted to learn, raw lap time included circuit-level variation that swamped the wear effect, and using the absolute first lap of each stint as a baseline was contaminated by cold-tyre warmup. After five iterations, we settled on WearDelta: the fuel-corrected lap time minus the fastest fuel-corrected lap in the warmup window ($\text{TyreLife} \leq 5$) of that stint. This isolates pure tyre wear by stripping out two confounders simultaneously, the ~ 0.03 s/lap pace gain from fuel burn-off, and the cold-tyre warmup penalty that biased earlier baselines.

Iteration	Target	Result
1	Lap time minus stint median	Negative R^2
2	Raw lap time	RMSE ~ 5 s; circuit-level variation swamps wear signal
3	Lap time minus first lap of stint	Negative target mean
4	Fuel-corrected lap minus first lap of stint	Cold-tyre warmup biases baseline
5	WearDelta: fuel-corrected lap minus fastest fuel-corrected lap in warmup window ($\text{TyreLife} \leq 5$)	Worked, final modeling target

We also engineered supporting features used by the optimizer: FuelCorrectedLap, TyreLifeRatio (tyre age normalized 0–1 within stint), CompoundEncoded (ordinal $\text{SOFT}=0$, $\text{MEDIUM}=1$, $\text{HARD}=2$), LapTimeDelta, and StintLength.

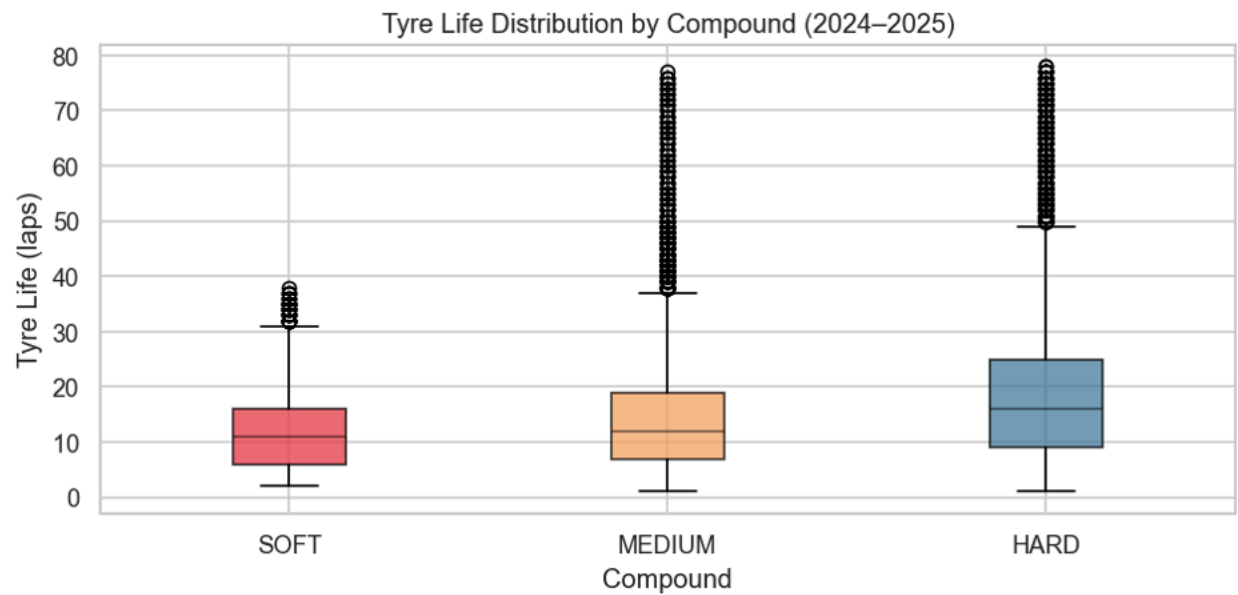


Fig 1: Tyre life distribution by compound

The compound-level stint distributions match expectation: SOFT runs short (median ~10 laps), HARD runs longest, MEDIUM has the widest spread reflecting its dual use as both short and long-stint compound.

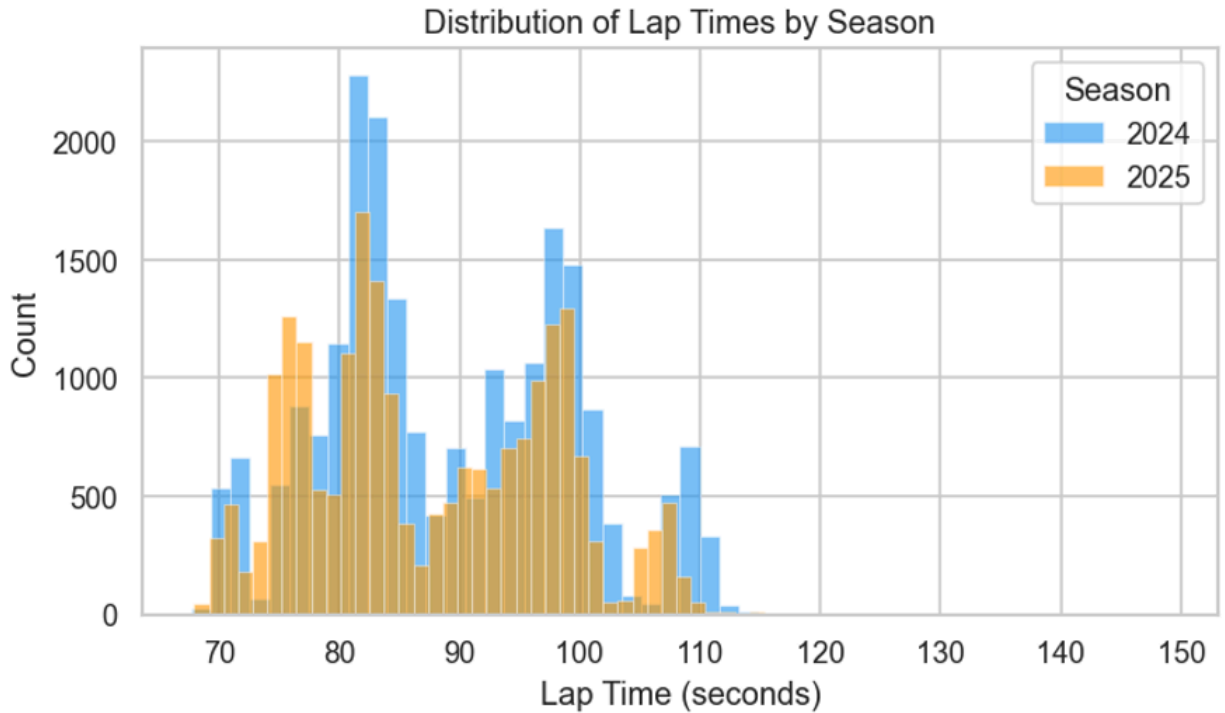


Fig 2: Distribution of lap times across the 2024 and 2025 seasons

In figure 2, we see lap time distributions for 2024 (blue) and 2025 (orange). Both seasons have a similar shape, with peaks reflecting different circuit lengths. But just because the raw lap times look alike doesn't mean the underlying tyre wear behaves the same way — Pirelli changed compound specs in 2025, and we found a real difference in how tyres degraded between seasons, which is why we mixed both years in our train/test split instead of training on one and testing on the other.

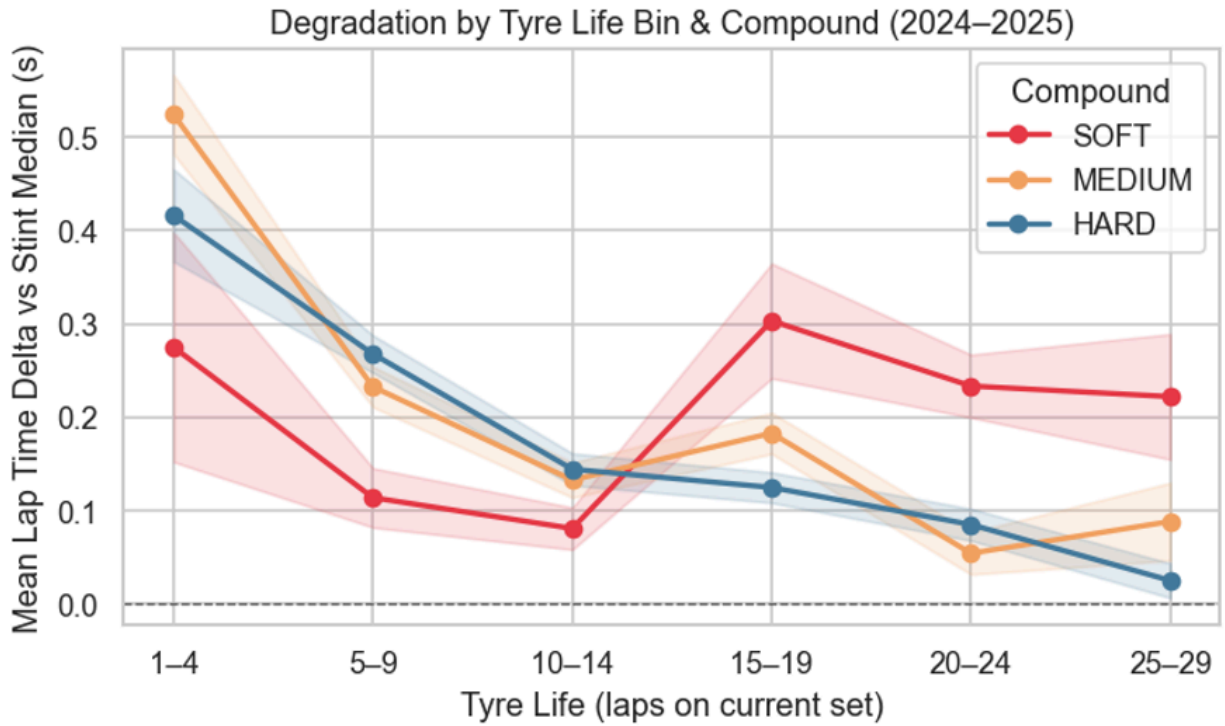


Fig 3: Mean lap-time delta relative to stint median as a function of tyre life and compound

The degradation by tyre life graphic shows the expected compound ordering: SOFT degrades fastest, HARD slowest, MEDIUM in between.

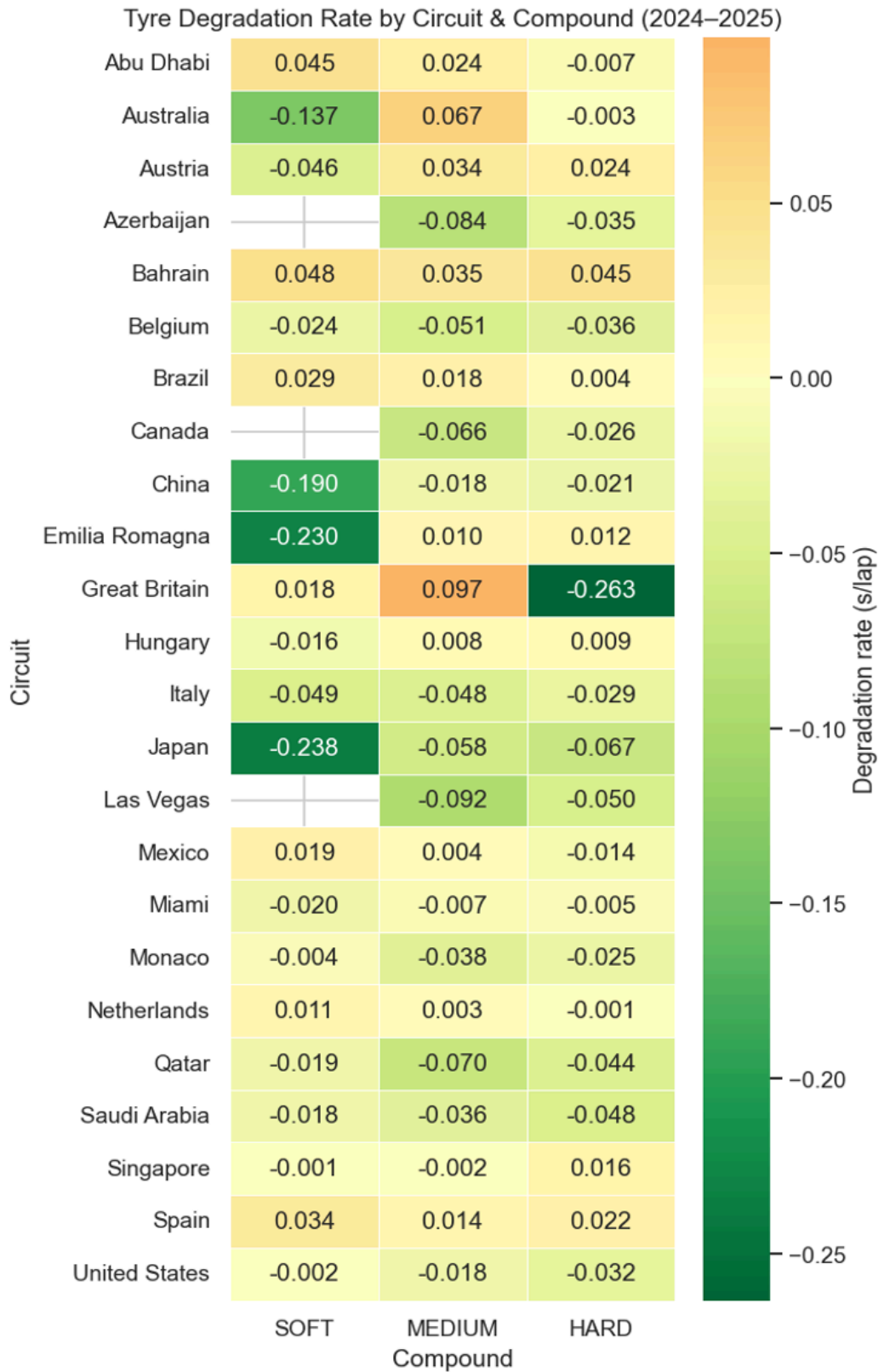


Fig 4: Degradation rate per circuit as a function of compound type

The heatmap quantifies a clear pattern, SOFT tyres almost always wear fastest at every track (magnitude of degradation varies significantly by circuit), followed by MEDIUM, and HARD tyres. High-grip, high-cornering tracks (Monaco, Hungary, Singapore) degrade tyres aggressively across all compounds; low-stress tracks (Monza, Spa) hold pace longer. This circuit-level variation is one reason our model captures only ~32% of WearDelta variance — a global model averages across these very different wear regimes. A per-circuit model is out of scope on this project, but is on our future-work list.

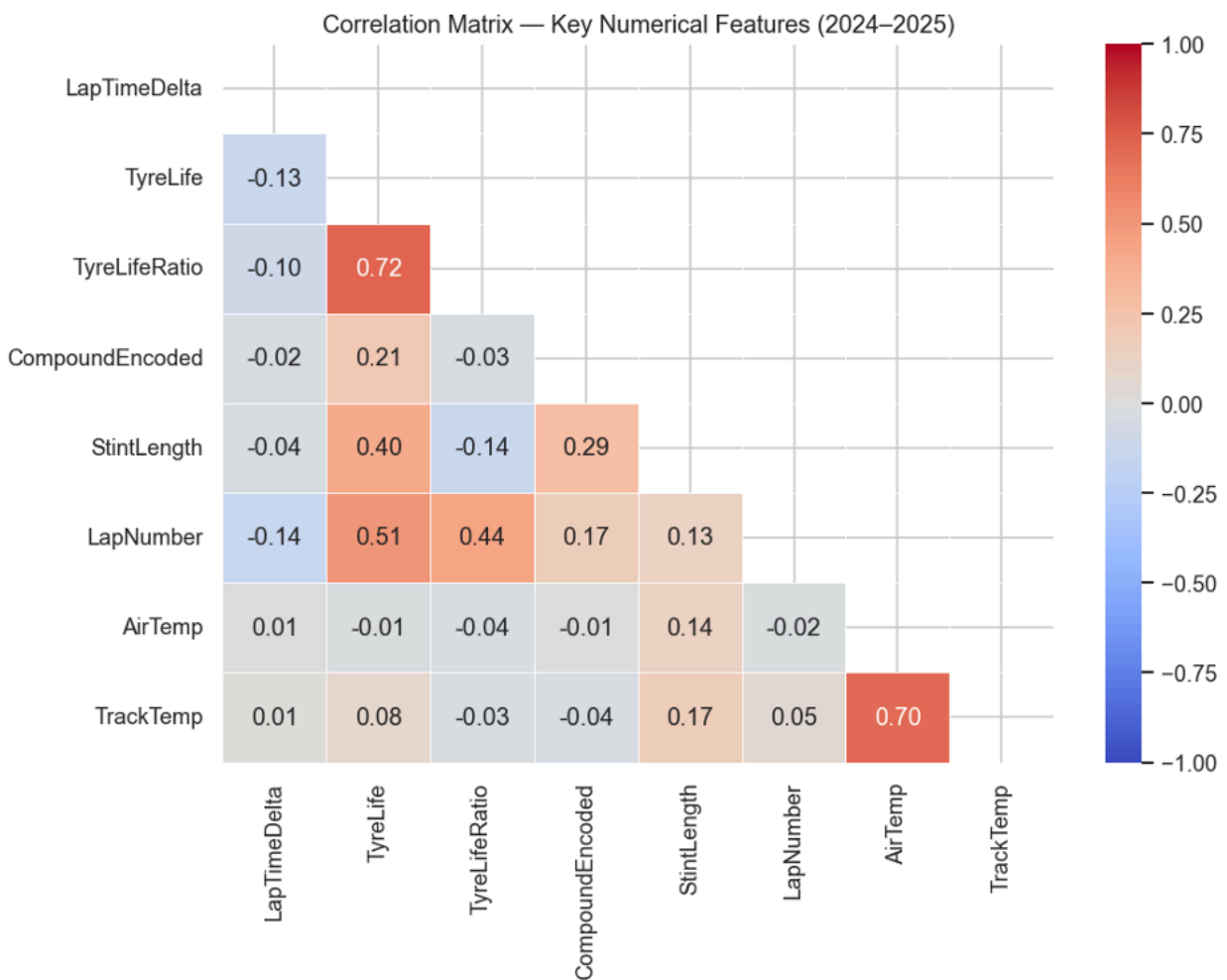


Fig 5: Feature correlation matrix

The correlation matrix highlighted that linear correlations with our target are weak (TyreLife → LapTimeDelta is just $r = 0.13$, the strongest single predictor). Compound type alone barely registers ($r \approx 0.03$), which makes sense. Its effect operates indirectly through stint length and tyre age. We also identified two pairs of highly redundant features (TyreLife/TyreLifeRatio at $r = 0.73$ and AirTemp/TrackTemp at $r = 0.70$) and dropped one from each pair. These weak linear signals signaled we'd need tree-based methods to capture more accurately interactions.

3. Methodology

We compared three regression models predicting WearDelta: Linear Regression (interpretable baseline), Random Forest (non-linear, ensemble), and XGBoost (gradient-boosted trees). Features were TyreLife, CompoundEncoded, TrackTemp, LapNumber, plus one-hot encoded Race.

The validation strategy required a notable pivot. Our initial plan, as stated in the proposal, was to train on 2024 and test on 2025, but we discovered Pirelli intentionally changed compound specifications between seasons to reduce degradation. The mean WearDelta shifted from +0.58 s/lap in 2024 to -0.07 s/lap in 2025. A 2024-trained model systematically over predicts on 2025 simply because the rubber is different. That makes the time-based split test the model's ability to predict regulation changes rather than its ability to predict wear from features. We pivoted to an 80/20 split at the stint level, mixing both seasons in train and test. Splitting at the stint level (rather than the lap level) matters because lap-level random splits would put laps from the same tyre stint on both sides of the split, which basically let the model "see the answer", splitting whole stints prevents that leakage. Two features were deliberately excluded: StintLength, which is only known after a stint ends and would constitute data leakage during prediction, and per-driver one-hot encodings, since driver-level pace does not transfer cleanly across team changes and seasons.

Once we had a wear model, we built a pit-stop optimizer on top. For each candidate strategy (a sequence of compounds with stint lengths summing to the race distance), the optimizer:

1. Looks up the fresh-tyre pace for each compound at that circuit
2. Predicts per-lap wear with XGBoost
3. Re-adds fuel weight
4. Adds a running on worn tyres incidental tax ($0.20 \text{ s/lap} \times \text{tyre age}$)
5. Adds 25 seconds per pit stop
6. Sums to a total predicted race time. It then searches every plausible 1 to 3 stop strategies, capped by per-compound stint maximums observed in the data, and returns the minimum.

The incidental tax was a key engineering choice. Without it, the optimizer always recommended 1-stop strategies with the longest possible first stint, because the wear model alone underestimates the cost of running long on worn tyres in real races (traffic, dirty air, defending position, lift-and-coast, none of which are in our features). The 0.20

coefficient was tuned against the 2025 back-test. We included it in the cost function specifically so the optimizer's recommendations would reflect race-day reality rather than pure pace optimization in a vacuum.

As an example, for Bahrain the optimizer recommended a 2-stop strategy: MEDIUM (20 laps) → HARD (16 laps) → MEDIUM (21 laps), with a predicted total race time of 94.6 minutes.

4. Results

Model	Train RMSE	Test RMSE	Test MAE	Test R ²
Linear Regression	1.60 s	1.58 s	1.04 s	0.13
Random Forest	1.30 s	1.48 s	0.93 s	0.24
XGBoost	1.09 s	1.40 s	0.89 s	0.32

Fig 6: Model performance comparison

XGBoost won the model comparison: Test RMSE = 1.40 s/lap, Test R² = 0.32. Random Forest came in second (RMSE 1.48, R² 0.24) and Linear Regression last (1.58, 0.13). Tree models clearly outperformed the linear baseline, confirming that the underlying wear behaviour is non-linear. Train/test RMSE gaps were modest, indicating the model isn't memorizing the training set.

R² of 0.32 means we explain about a third of the variation in lap-time wear within a stint. That sounds modest but given everything else happening in an F1 race that has nothing to do with tyres (traffic, dirty air, fuel saving, drivers managing pace on engineer instruction) a third of the variance is a meaningful share. The proposal's RMSE < 0.3 s/lap target turned out to be unreachable: F1 lap times have several seconds of irreducible non-tyre noise, and no per-lap regression model without telemetry features can get below that floor.

The optimizer back-test on the 20 dry 2025 races (we excluded Belgium, Canada, and Great Britain as wet/chaos races) tells an interesting story. Compared against just the race winner, we matched stop count and compound mix on 55% of races, with predicted race times within ± 120 seconds of the winner's actual race time on 60%. The race-winner comparison is noisy, though the winner's strategy is shaped by qualifying position, available tyre allocation, and in-race events the optimizer doesn't see. The more honest test is whether the optimizer recommends a strategy that any

points-scoring (top 10) finisher actually used. By that measure we hit 80% on stop count and 85% on compound mix. When the optimizer makes a recommendation, some competitive driver did execute that strategy on 16 to 17 races out of 20. By our original success criterion, a strategist can look at this tool's output and call it sensible. The approach worked.

5. Limitations and future work

Several limitations bound the current results. Per-lap noise from traffic and driver inputs sets an irreducible RMSE floor that no model without telemetry features can break through. The 0.03 s/lap fuel correction is a global rule of thumb when the actual fuel effect varies by circuit and car. Pirelli's compound spec changes between seasons mean our model averages two different rubber generations rather than specializing on either. The 0.20 incidental tax coefficient was hand-tuned against the back-test we evaluate on, which isn't fully separated.

Given more time, the highest-impact extensions would be:

1. per-circuit wear models, fitting separate degradation curves per circuit-compound pair, closer to how F1 strategy engineers actually work
2. per-lap weather features instead of race-mean track temperature, since track conditions can swing significantly during a race
3. per-circuit fuel correction to replace the global 0.03 s/lap constant.

We'd also hold out a separate calibration set for the incidental tax to make the back-test fully honest end-to-end.